PS 3: Week 12

Multiple Linear Regression and Experiments

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Overview

Attendance Survey

Fill out before the end of section!

- Attendance survey form (forms.gle/88qgrMWiEHheWFqc9)



Agenda

1. Attendance Survey
2. MLR: Experiments
3. MLR: Experiments: Worksheet
4. Interpreting Regression Tables
5. Regression Table Example: Observational

Housekeeping

- First Draft of Report due 11/21
  - You can look at results as a group

  - **Write the reports individually**
    * Three statistical tests + interpretations
- Office Hours for cleaning/testing the data

MLR: Experiments

Multiple Linear Regression + Experiments

What do experiments help us achieve?

- Randomization lets us identify a causal effect by eliminating sources of **bias**
  - Because treatment is randomized, we cannot have omitted variable bias
  - Because we assign treatment before measuring our outcome of interest, we cannot have reverse causation
  - Because we—and not the participants—decide whether someone is assigned to treatment, we cannot have selection bias
- Therefore, the treatment effect we see in an experiment can be seen as causal
  - Ex: The treatment causes a 5-point increase in voting compared to the control

Multiple Linear Regression + Experiments

What does multiple linear regression help us achieve?

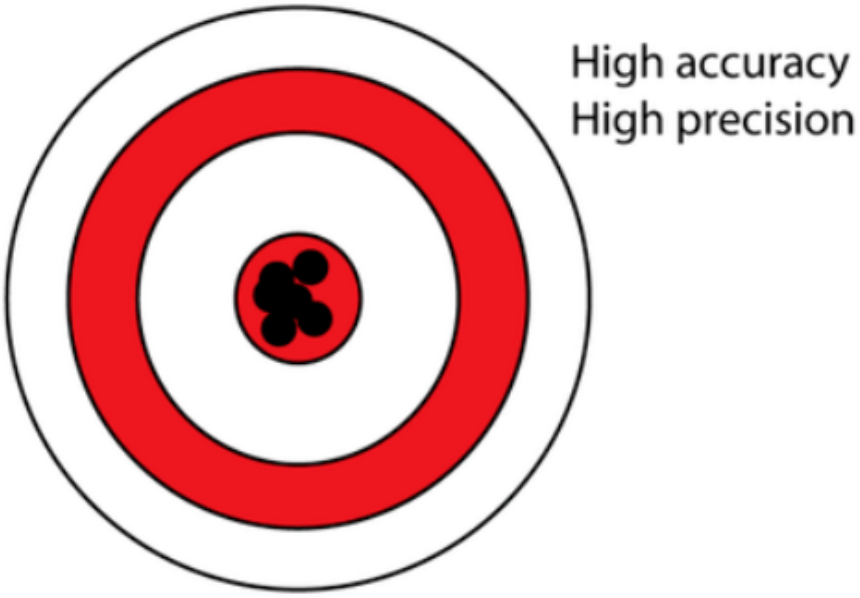
- Beyond bias, we are worried about **noise**
  - Recall that randomization of treatment creates noise
    * *Did we observe our results do to random chance?*
- We can reduce noise by performing a multiple linear regression
  - This will allow us to look at the treatment effect while controlling for other variables that may be related to the outcome, such as demographics or study conditions
- The treatment effect is still the **only causal effect we identify**
  - The estimates we find for the other variables in our regression are only associations or correlations, not causal effects
    * **No causation without randomization!!!!!!!!!!**

No Bias, Minimal Noise

Randomized Experiments + Multiple Linear Regression

No bias + Minimal noise

This gives us an **unbiased** and **precise** estimate of the true average treatment effect in the population



MLR: Experiments: Worksheet

Regression Worksheet (Experiments)

Table 1: Relationship Between Democratic US House Vote Share in 2020 and Campaign Spending in 2020, With Controls

	Democratic Incumbents	Republican Incumbents
Democratic Spending $ millions, 2020	-0.216 (0.185)	
Republican Spending $ millions, 2020		0.138 (0.161)
Clinton Vote Percent, 2016	0.789*** (0.074)	0.800*** (0.071)
Democratic US House Percentage, 2018	0.107 (0.065)	0.201*** (0.053)
Constant	6.156 (3.223)	-2.684 (2.699)
N	210	191
R^2	0.6707	0.5276

Notes: The outcome of both regressions is the percentage of the vote that Democrats obtained in US House elections in 2020. Each observation is one Congressional district. The column headings show the subset of the data in which the regression was run. $*** = p < 0.001$, $** = p < 0.01$, $* = p < 0.05$.

Interpreting Regression Tables

Reading a Regression Table

Each **column** reports the results of one regression. Multiple columns means we are looking at the results of multiple different regressions (in this example, two). The **column titles** should tell you what is different about the regressions.

- Sometimes this will be switching subsets of data (as in this example).
- Sometimes it will involve switching *outcomes*.
- Sometimes the outcome and subset will be the same, but new variables will be introduced into the regression.
- You have to use context and the table notes to be sure what is changing across the columns.

Reading a Regression Table

Each **row** represents the coefficients/estimates on one of the variables in the regression.

- The **first column** shows the variable that is being estimated.
- The **last row** of estimates contains the Intercept or Constant term.

In each **cell**, the first number is the coefficient, and the number in brackets below it is the standard error. These are the only numbers we need to calculate a t-statistic, confidence interval, and p-value.

The **stars** next to each number indicate if the estimates are statistically significant and, if so, how small the p-value is. (The **Table Notes** along the bottom will tell you how to interpret the stars.)

To re-cap

Missing cells mean that this variable was not included in this regression. Sometimes it will just be blank, sometimes there will be a "-" to indicate it is being left blank on purpose.

The **sample size** and **R squared** are printed at the bottom of each column.

- R^2 or R squared is the proportion of the variation in the dependent variable that is predictable from the independent variable(s).

Steps for Interpretation!

- Look at the **title** of the table
  - What are the authors generally trying to tell us?
  - This can clue you into relevant variables and takeaways, though keep a critical eye to the table yourself!
- Look at the title of each **column** (this is the title of each regression model)
  - What is each model trying to tell me? What **subset** of data are they using? What **outcome**?
- Look at the **row** labels (these are the predictor variables)
  - Which ones are **causal** (treatment assignment)? Which ones are **not** (any other variables)?
- Look at the point **estimates** themselves
  - Based on the row label and column title, what is this estimate telling me?
- Look at the measures of **noise**
  - Is the estimate **significant**? How do I interpret significance in this context?
    - Is it causal?

Example Table: Table Title

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Example Table: Column Titles

Table 1: Relationship Between Democratic US House Vote Share in 2020 and Campaign Spending in 2020, With Controls

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Example Table: Row Labels

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Example Table: Point Estimate

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Example Table: Significance Stars

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Regression Table Example: Observational

Example

Top-left box of the previous table:

- Based on the title, it seems that we're supposed to look at how Democratic spending is related to Democratic vote share in 2020 Congressional races. "Controls" implies a multiple linear regression.
- The model is called "Democratic incumbents". The outcome is the vote share for Democrats in 2020 Congressional races, but only in races where Democrats already held the seat.
- The row refers to Democratic spending in 2020 and should be interpreted in millions of dollars.

Example (continued)

- Given that "vote share" is in percentage points, and Democratic spending is in millions of dollars: an increase of $1 million dollars in Democratic spending is associated with, on average, a 0.22% decrease in Democratic vote share in 2020 Congressional races where Democrats were the incumbents.
- This relationship is not significant (note that the standard error and estimate are of similar size!)
  - **None of these variables have a causal interpretation because the variables are not randomly assigned.**

Wrap-Up

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Acknowledgements

Special thanks to prior GSIs for providing materials off of which some of these slides are based, especially Yue Lin.